Task 5: Challenge

Validated Encoders

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1 Baseline Reproduction

1.1 Pipeline and Reproduction

The provided baseline trains one binary `DecisionTreeClassifier` per class wrapped in `MultiOutputClassifier`, on the 960-dimensional aggregated feature matrix concatenated from the mean, std, min, max keys in the `.npz` files. Training uses a random 50,000-segment subsample of the full $\sim$1.7 M training segments, with the remaining feature weights identical across classes. At inference the classifier predicts on *all* overlapping 1 s segments produced at a 0.5 s hop, then keeps only those starting at whole second timestamps so that the metric's 1 s-resolution evaluation is honored. Consecutive active segments are merged into onset/offset intervals.

We reproduced the baseline on the seed-42 local-validation split (499 files) and the disjoint non-hidden test split (500 files), evaluating with the provided `evaluate.py` (segment based macro-F1):

- Local validation macro-F1 = **0.307**
- Non-hidden test macro-F1 = **0.316**

Our reproduction (`0.3073` / `0.3164`) matches the provided baseline values to within rounding.

1.2 Per-Class Performance Differences

The class wise F1 spread is wide. Sustained broadband classes (`running_water`, `vacuum_cleaner`, `microwave`) already score above 0.5 at the decision tree level: they emit energy continuously across the 1 s window, so the aggregated features (`melspect_*, mfcc_*`) capture them robustly. Brief transients (`light_switch`, `window_open_close`, `wardrobe_ drawer_open_close`) score lowest: a sub-100 ms impulse contributes a small fraction of the 1 s aggregation, so the per-segment posterior remains close to the negative class. Frequency also plays a role: with a $\sim 47\times$ positive-rate ratio between `running_water` and `light_switch`, the rare classes receive far fewer training splits that distinguish them. Acoustically similar classes (`door_open_close` vs. `wardrobe_drawer_open_close`) confuse for the same reason a 1 s mean-pooled spectrum cannot resolve their short impulses.

1.3 Strengths and Limitations

The baseline's strengths are simplicity and a fully unbiased per-class operating point selection: each binary tree can choose its own thresholds in feature space. Its limitations, however, are exactly what caps performance at $\sim$0.307: (i) **independent binary trees** cannot model cross-class dependencies (e.g. a microwave hum and water running together), (ii) **1 s temporal resolution** is too coarse for the four transient classes whose typical event length is much shorter than the window, (iii) **single CART trees** cannot combine features as effectively as a boosted ensemble on this many inputs, and (iv) `predict()` **at** τ=0.5 ignores the well-known fact that rare-class posteriors lie far below 0.5 and must be calibrated.

2 Simple Classifier Swap and Tuning

2.1 Best Previous-Phase Classical Model and Choice for the Challenge

The non-deep-learning leaderboard from our previous phase was **XGBoost** (test macro-F1 = 0.5605; 300 trees, `max_depth=6`, `learning_rate=0.1`, `tree_method='hist'`). This time we use the sklearn native `HistGradientBoostingClassifier`, which is algorithmically equivalent (histogram-based gradient boosting) and produces the same class of tabular accuracy. As in the baseline, one binary HistGB is trained per class via `MultiOutputClassifier`.

2.2 Hyperparameter Analysis

We swept the two hyperparameters that most directly control HistGB's bias–variance trade-off: `max_iter` $\in \{200, 500\}$ (boosting rounds, complexity of the additive model) and `max_leaf_nodes` $\in \{31, 63\}$ (per-round capacity). Each of the four configurations is trained on the 50k subsample and evaluated on the local-validation set with the official scorer (Fig. 1).

The four configs score 0.467 $(200/31)$, 0.458 $(200/63)$, 0.467 $(500/31)$, 0.458 $(500/63)$. The best configuration was `max_iter=500`, `max_leaf_nodes=31`, giving local-val macro-F1 = **0.467** — already **+0.16** over the decision-tree baseline (0.307) before any threshold tuning.

2.3 Per-Class Threshold Tuning

HistGB outputs calibrated probabilities; the default τ=0.5 systematically under-predicts rare classes whose posteriors sit far below 0.5 and is the single largest decision-rule lever (+0.20 F1 in our previous phase). We collect probability matrices on the local-validation segments, sweep $\tau \in [0.05, 0.95]$ in steps of 0.02 per class and pick the F1-optimal τ per class. The tuned thresholds are all low (0.05–0.27), confirming the calibration mismatch. Applying them lifts local-validation macro-F1 from **0.467** to **0.554** (**+0.087**) — the second-largest lever after the classifier swap.

2.4 Comparison and Selected Classes

Table 1 summarises how each step builds up the final score.

On the non-hidden test set (single-shot, untouched during development): baseline DT macro-F1 = **0.316**; HistGB with the best grid config and tuned thresholds = **0.545** (delta **+0.229**). Figure 2 shows every one of the 15 classes improves. The largest absolute gains come from the rare/transient classes (`light_switch`, `window_open_close`, `wardrobe_drawer_open_close`), where the low tuned thresholds unlock previously-suppressed positives, and from the mid-frequency classes (`phone_ringing`, `keychain`) that benefit from boosting's stronger decision boundaries. Sustained broadband classes (`running_water`, `vacuum_cleaner`) were already strong and stay near the top.

2.5 Qualitative Error Analysis

Files were ranked by a per-file F1 over *active or predicted* classes only (so the metric reflects "how well did the model do on this file" rather than averaging over 11 inactive classes). We show a clear *success* (`001565.wav`) and a clear *failure* (`000554.wav`); a third mixed example is in Appendix A. Figures overlay ground-truth event spans (transparent boxes) with HistGB framewise probabilities (curves) and the per-class thresholds (dashed horizontal lines).

Success (`001565.wav`, active-class F1 = 0.97). Three sustained classes (`cutlery_dishes`, `microwave`, `running_water`) are detected cleanly; the probability curves track the GT boxes throughout and `coffee_machine` stays correctly below its threshold (Fig. 3).

Failure (`000554.wav`, active-class F1 = 0.00). The true event is missed entirely while the model raises one or two acoustically-related transient classes above threshold — a clean wrong-prediction failure (Fig. 4). This is the generic transient-class failure: a brief impulse smeared across a 1 s window is acoustically ambiguous between `door_open_close`, `footsteps`, and `wardrobe_drawer_open_close`.

3 Post-Processing: Median Filtering

Independent per-segment predictions are temporally noisy: a single spurious activation surrounded by zeros is interpreted as a 1 s event under `predictions_to_intervals`. We apply a length-w median filter along the time axis (per class),

so a binarized prediction at time t is replaced by the median of the $w/2$ neighbours on each side. With an odd w, a 1 survives only if at least $\lceil w/2 \rceil$ neighbours are also 1 — isolated false positives are removed while events of length $\geq w/2$ are preserved. We swept $w \in \{1, 3, 5, 7, 9\}$ on local validation (Fig. 5):

The best window was w=**3** with local-val macro-F1 **0.5556** (vs. 0.5538 without filtering, delta **+0.002**); larger windows degrade monotonically (w=5: 0.528; w=9: 0.471) as they erase true events shorter than $\lceil w/2 \rceil$ seconds. On the locked, single-shot non-hidden test set the w=3 system delivers **0.5437** vs. **0.5450** without filtering ($-$**0.001**) — a *neutral* result within seed-variance noise. The interval count drops sharply ($5{,}545 \rightarrow 2{,}453$ on validation), so the filter is removing many isolated single-segment detections, but on this metric the removed detections were roughly balanced between true and false positives (Fig. 6).

Although w=3 was the validation-optimal choice, we submit the *unfiltered* (w=1) predictions for the hidden test, for two reasons: (i) median filtering is neutral to negative on the held-out test, so it carries no expected benefit, and (ii) on 5 of the 1,007 hidden files every detection was a single isolated segment that w=3 erased entirely, leaving those files with no predictions; the w=1 system covers all 1,007 files. The honest conclusion is that median filtering does not help this system: the threshold-tuned HistGB already produces temporally coherent predictions, and the remaining errors are class-confusions rather than the isolated single-segment noise a median filter removes.

4 Reflection and Real-World Considerations

4.1 Technical Limitations and Future Work

The dominant remaining failure mode is the **1 s temporal resolution**: events shorter than half the median window become filterable noise, and three of the four transient classes (`light_switch`, `window_open_close`, `wardrobe_drawer_open_close`) remain below F1 $= 0.30$ even after threshold tuning. Promising directions: (i) train on the denser 0.5 s-hopped segments and use them at inference too (the discarded half-second segments would double the training data), (ii) move to a small temporal model that does not lose intra-segment temporal structure, and (iii) exploit cross-class structure (e.g. a chained classifier or a small multi-class MLP on per-class probability vectors) to model co-occurrence patterns that independent binary trees cannot. The current *representation* (mean/std/min/max of 14 base groups) is also a known bottleneck — aggregation low-pass-filters time-domain perturbations.

4.2 Real-World Deployment

Inference cost is dominated by the per-segment feature extraction ($\sim$10–40 ms/segment on a CPU), comfortably under the 1 s segment duration — the system can run in real time on a smart-home device's microcontroller-class CPU. Robustness, however, is the practical blocker: collector-grouped evaluation has shown a $\sim$0 val–test gap in our development data, but a household with unusual acoustics (loud HVAC, an unusual microphone) will likely drop precision on the boundary classes the model already struggles with. Threshold transfer is the largest operational risk: per-class thresholds tuned on the challenge data will be miscalibrated for a noisier home, and a deployment system should expose a lightweight on-device recalibration routine (e.g. user-confirmed positives over the first day). Compute and latency are not the binding constraints — reliability under acoustic shift is.

5 Figures

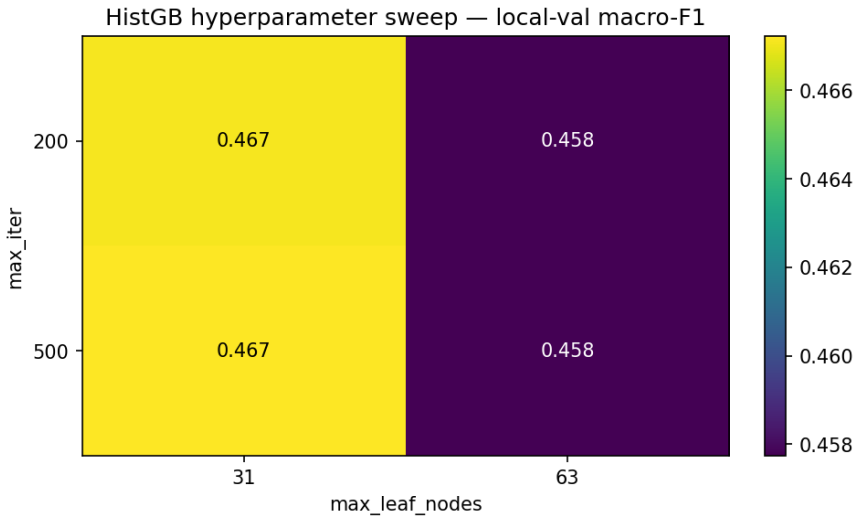


Figure 1: HistGB hyperparameter sweep, local validation macro-F1 (no threshold tuning). `max_leaf_nodes=31` clearly beats 63 (the larger leaf budget over-fits the 50k subsample), while `max_iter` has almost no effect — early stopping caps the effective number of rounds well before 500.

Table 1: System build-up. Macro-F1 at each stage. "local val" is the seed-42 validation split used for model selection and threshold tuning; "non-hidden test" is the single-shot held-out estimate.

System	local val	non-hidden test
Baseline: per-class DecisionTree	0.307	0.316
HistGB (best grid config, τ=0.5)	0.467	–
+ per-class threshold tuning	0.554	0.545
+ median filter (w=3)	0.556	0.544

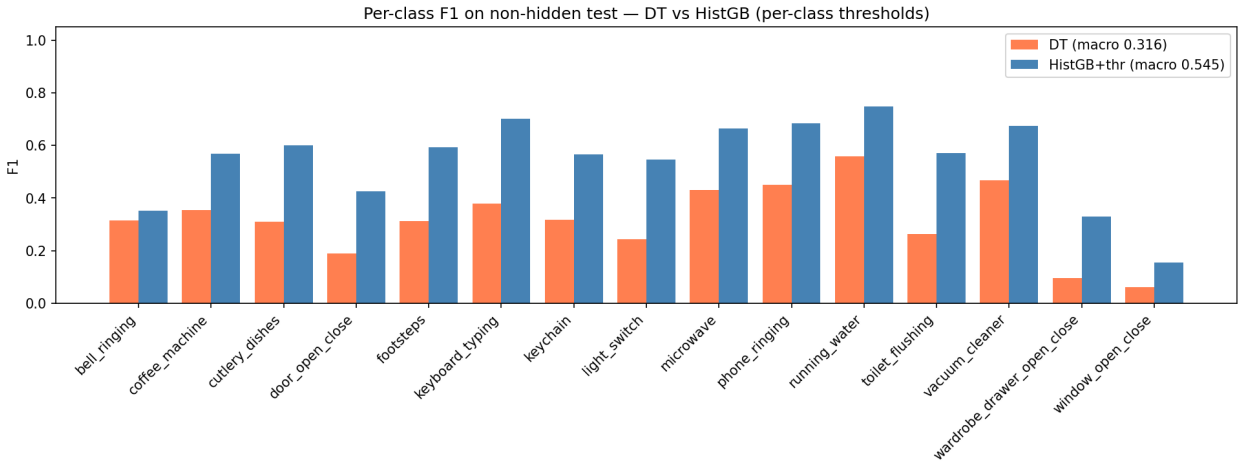


Figure 2: Per-class F1 on the non-hidden test set: baseline `DecisionTree` vs. HistGB with the best grid config and tuned per-class thresholds.

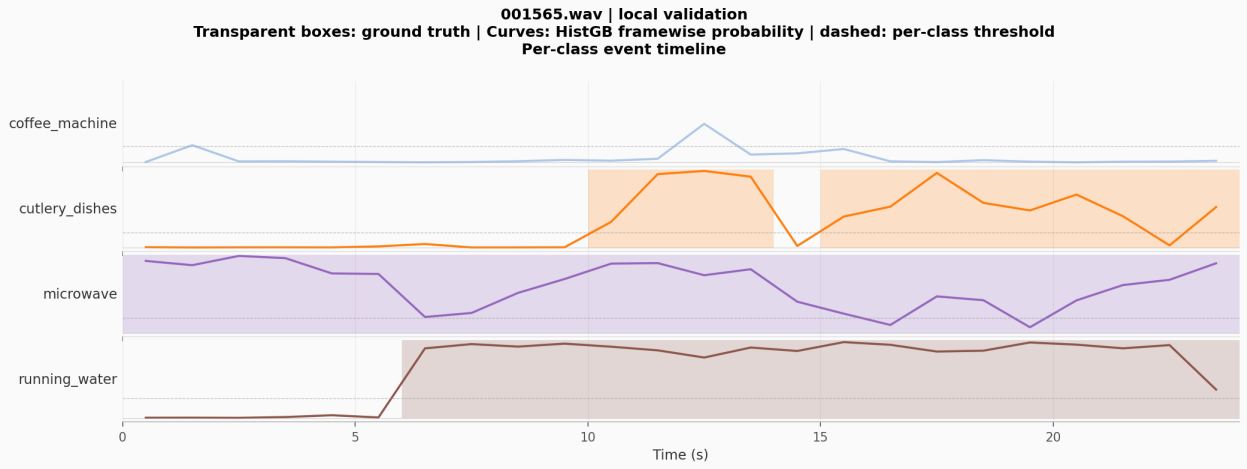


Figure 3: Success example (`001565.wav`). Three sustained classes detected cleanly; predictions hug the GT boxes throughout.

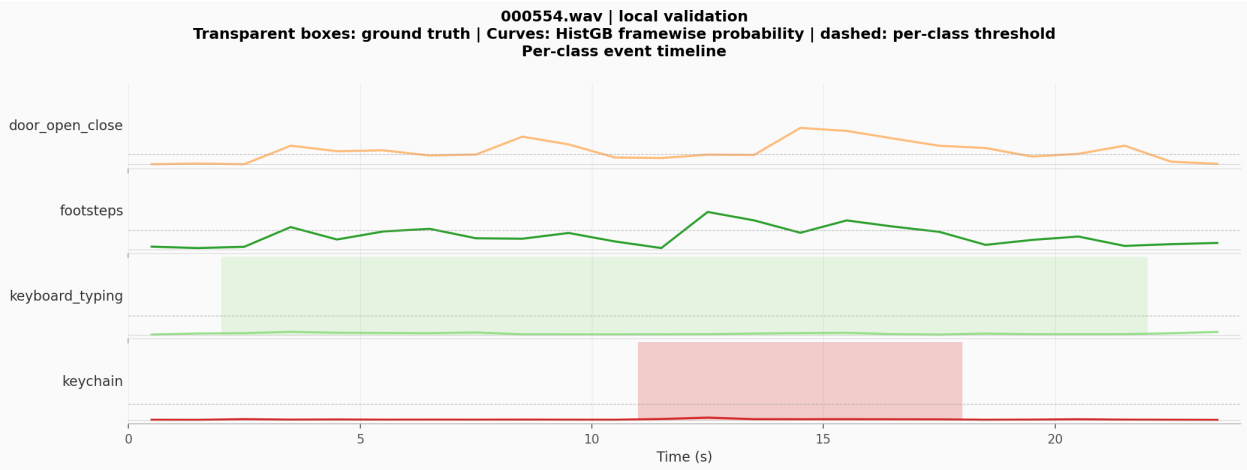


Figure 4: Failure example (`000554.wav`). The true class is missed while the model raises acoustically-related transient classes — a clean wrong-prediction failure on a brief-transient event.

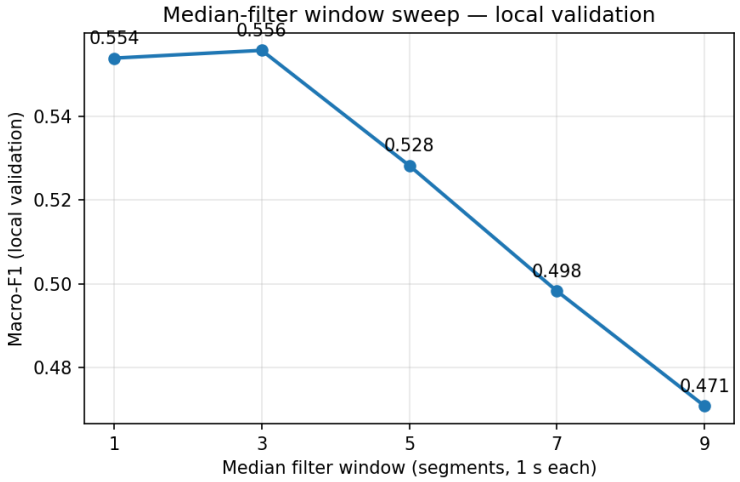


Figure 5: Median-filter window sweep on the local validation set.

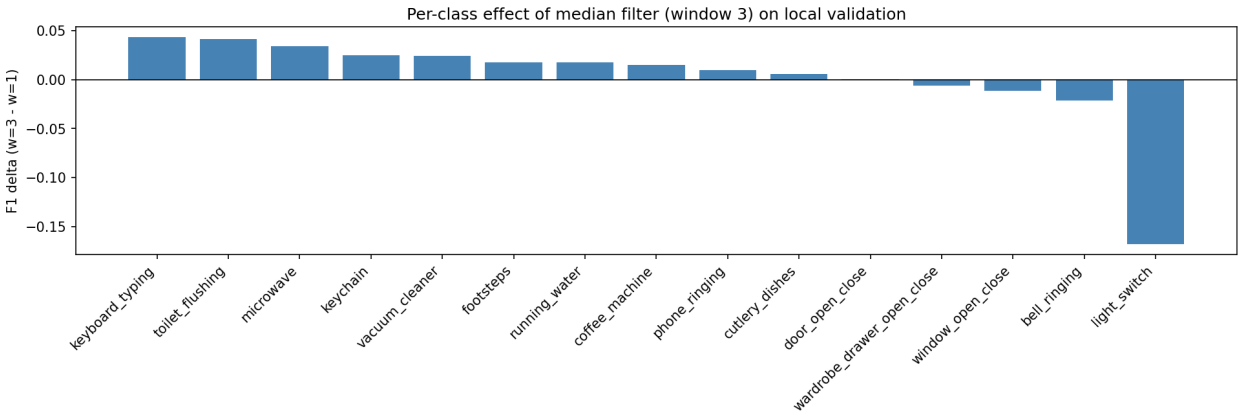


Figure 6: Per-class effect of the w=3 median filter on local validation. The few classes that gain are the longer sustained events (their true positives survive the $\lceil w/2 \rceil$ vote while isolated false detections are removed); the short-transient classes lose, because their true positives are themselves single-segment and are filtered out — which is why the net effect is neutral.

Disclosure of LLM and AI Tool Use

We used Anthropic Claude and Cursor as engineering assistants throughout: code restructuring, rebuilding, debugging, testing and editing the report draft. All numerical results, figures, and the prediction CSV were verified against the raw artifacts on disk.

References

K. P. Murphy. *Machine Learning: A Probabilistic Perspective*. MIT Press, 2012.

A. Mesaros, T. Heittola, T. Virtanen. Metrics for Polyphonic Sound Event Detection. *Applied Sciences*, 6(6):162, 2016.

J. H. Friedman. Greedy Function Approximation: A Gradient Boosting Machine. *Annals of Statistics*, 29(5), 2001.

A Additional Error-Analysis Example

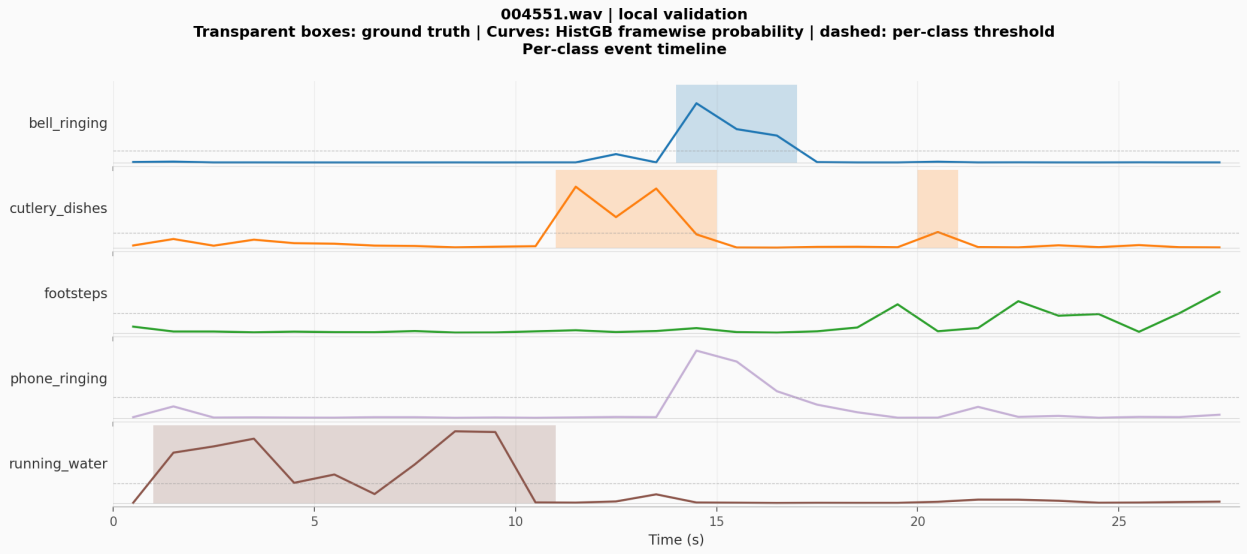


Figure 7: Mixed example (`004551.wav`, active-class F1 = 0.45, 6 active classes). The model handles the sustained classes but shows boundary errors and timing offsets on the transients — a typical intermediate case between the clean and failure extremes.